

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and deployed **machine learning classification models** using **scikit-learn, pandas, and NumPy** on production datasets of 500K+ transaction records, achieving **94% accuracy** in predictive analytics and reducing manual data validation effort by **40%**.
- Built end-to-end **data pipelines** and **ETL workflows** using **Python** and **SQL (PostgreSQL, MySQL, Oracle)** to integrate multi-source enterprise data from **ERP systems** and operational platforms, enabling real-time analytics dashboards that decreased reporting latency by **65%**.
- Developed **interactive Power BI dashboards** and **Tableau visualizations** using **DAX measures** and **data modelling** to track KPIs across financial services operations; published reports to 50+ stakeholders, driving **30% improvement** in decision-making cycle time.
- Performed **statistical analysis, hypothesis testing, and A/B testing** on payment transaction flows and banking operations data using **R and RStudio**; delivered predictive models for payments reconciliation that reduced settlement discrepancies by **28%**.
- Led **data warehousing design** and schema optimization across **AWS (EC2, S3, Lambda)** using **CI/CD pipelines (Jenkins, Git)** to operationalize analytics models; automated scheduled data loads via **cron jobs** and **Linux shell scripts**, reducing manual extraction overhead by **55%**.
- Managed **big-data analytics** projects leveraging **Hadoop and Spark** concepts to process 2TB+ datasets; collaborated with cross-functional teams on exploratory data analysis, time-series forecasting, and feature engineering to support regulatory compliance and GDPR data-protection audits.

SKILLS

Data Science & Machine Learning – scikit-learn, supervised learning, unsupervised learning, classification, regression, model evaluation, feature engineering, hypothesis testing, A/B testing, predictive analytics.

Data Analytics & Visualization – Power BI, DAX, Power Query, Tableau, data modelling, interactive dashboards, business intelligence, statistical analysis, exploratory data analysis.

Data Engineering & ETL – Python, pandas, NumPy, SQL, data pipelines, ETL workflows, data warehousing, PostgreSQL, MySQL, Oracle, schema design, data integration.

Big Data & Distributed Computing – Hadoop, Apache Spark, large-scale data processing, distributed analytics.

Programming & Infrastructure – Python, Java Spring Boot, R, RStudio, AWS (EC2, S3, Lambda), Docker, Linux, shell scripting, cron jobs, Git, Jenkins, CI/CD pipelines.

Statistical Methods & Analytics – time-series forecasting, regression analysis, statistical testing, data validation, quantitative modelling.

Domain & Compliance – financial services, banking operations, payments processing, GDPR, data protection, regulatory compliance, ERP systems integration.

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

PROJECTS

Real-Time Payments Analytics & Predictive Fraud Detection Platform (HCL Technologies)

- Architected a **machine learning pipeline** using **scikit-learn** and **pandas** to detect anomalies in 10M+ daily payment transactions across SEPA and cross-border payment flows, achieving **96% precision** in fraud classification and reducing false-positive rate to **2.1%**.
- Built **data warehousing layer** on **AWS (S3, EC2, Lambda)** to consolidate ISO 20022 message flows and transaction metadata from multiple banking systems; implemented **automated ETL** using **Python** and **PostgreSQL** to process schema transformations with **99.8% uptime SLA**.
- Developed executive **Power BI dashboards** and **Tableau visualizations** with **DAX measures** tracking real-time payment settlement KPIs, transaction-flow visibility, and regulatory compliance metrics (GDPR audit trails, PSD2 data residency); reduced reporting generation time from 4 hours to **12 minutes**.
- Implemented **time-series forecasting models** using **R** and **statistical methods** to predict transaction volume spikes and payment-processing bottlenecks, enabling proactive capacity planning and reducing operational incidents by **35%**.
- Operationalized analytics models via **CI/CD pipelines (Jenkins, Git)** and **Linux cron-scheduled jobs** for continuous model retraining; deployed inference layer on **AWS Lambda** for sub-100ms latency prediction serving 500K transactions daily.

Cross-System ERP Data Integration & Business Intelligence Dashboard Suite (HCL Technologies)

- Designed and deployed **SQL data pipelines** across **PostgreSQL, MySQL, and Oracle** to unify enterprise business-system data from ERP modules, operational platforms, and financial applications; supported change-control management and UAT coordination, enabling stakeholders to access integrated reporting **2 weeks ahead of schedule**.
- Created multi-layer **data model** and **Power BI dashboards** with **Power Query ETL** to visualize cross-system reconciliation workflows and transaction lineage across interconnected ERP systems; delivered **12 interactive reports** to finance and operations teams, reducing manual data pulls by **60%**.
- Performed **exploratory data analysis** and **statistical validation** on ERP transaction records to identify data quality issues and missing schema relationships; implemented **automated data validation scripts** in **Python** to flag reconciliation discrepancies before post-go-live production support, reducing issue resolution time by **45%**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)